

1. PROJECT OVERVIEW

- **Project Title:** FitBuddy - AI Workout Generator

2. FUNCTIONAL REQUIREMENTS (FR) These represent the core operational capabilities the FitBuddy software must execute.

- **FR1 (Data Collection):** The application shall develop a responsive web interface to collect user fitness profiles (Age, Weight) and goals.
- **FR2 (Generative AI Integration):** The backend shall integrate the Google Gemini API to dynamically generate a customized, day-by-day workout plan based on the submitted data.
- **FR3 (Database Management):** The system shall utilize a relational database to persistently store user profiles and their corresponding AI-generated plans for future retrieval.
- **FR4 (Admin Dashboard):** The application shall provide an Admin UI to view all registered users and their workout plans in a tabular format.
- **FR5 (Exception Handling):** The backend shall implement seamless fallback mechanisms (such as timeout handling) to ensure a continuous user experience during API latency.

3. NON-FUNCTIONAL REQUIREMENTS (NFR)

These dictate the qualitative aspects of the application.

- **NFR1 (Framework Speed):** The application must be built on a fast and modern web framework (FastAPI) to ensure rapid routing and low-latency internal processing.
- **NFR2 (Reliability):** The AI generation module must handle network timeouts gracefully (e.g., setting a 60-second timeout parameter in the API call) to prevent server crashes.
- **NFR3 (Scalability):** The use of SQLAlchemy ORM must allow for easy future migration from SQLite to larger databases like PostgreSQL if user scaling is required.

